

Cloud & Workload Identity Migration

2026, 2027 and beyond

Version 2.0 | Reviewed 29 September 2026

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Static PDF companion

Use the matching XLSX for editable inventory records, formulas, drop-downs, evidence, owners and status. This PDF is a readable reference and is not an interactive workbook.

How to use this companion

Start with scope and accountable owners. Record the cryptographic dependency, confidentiality or authenticity lifetime, algorithm and implementation evidence. Prioritize exposed, long-lived, high-impact dependencies; confirm supported protocol and product paths; test interoperability, failure modes, rollback and operational impact before deployment.

Final standards, drafts, research selections and local planning targets are different statuses. NIST alignment means the material draws from the listed sources; it does not claim NIST endorsement or completeness of every future recommendation.

Workbook: [secureazcloud-pqc-cloud-workload-identity-checklist-v2.0.xlsx](#)

Included worksheets

[Start Here](#) | [Checklist](#) | [Workload Identity Inventory](#) | [Certificate and Key Map](#) | [CISA Category Map](#) | [Pilot Test Plan](#) | [Sources](#)

Start Here

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

Redacted public checklist for cloud, IAM, certificates, workload identity, token signing, PKI, TLS, SaaS, key lifecycle, and supplier-readiness migration planning.

Version: v2.0

Updated: 2026-09-29

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Purpose: Use this checklist to translate PQC readiness into cloud and identity implementation tasks across workload identity, managed certificates, federation, KMS/HSM, SaaS dependencies, and CI/CD signing.

NIST basis: FIPS 203/204/205, SP 800-227, CSWP 39 Update 1, IR 8587, SP 800-228 Update 1 and NCCoE migration resources. Product-family mapping is supplementary.

How to use: Work through the Checklist sheet, collect evidence, then populate Workload_Identity_Inventory and Certificate_and_Key_Map for P1/P2 assets.

Evidence standard: Use architecture diagrams, IAM exports, certificate scans, KMS/HSM reports, service mesh configs, CI/CD signing settings, vendor roadmaps, and pilot test records.

Caution: This is a planning tool. Do not enable production PQC or hybrid cryptography without vendor support, protocol support, interoperability testing, observability, and rollback approval.

2026 baseline: Assign owners, discover public-key dependencies, assess data and signature lifetimes, confirm supplier versions and establish tested recovery.

2027 delivery: Pilot supported profiles, verify relying parties and operational limits, approve migration waves and require measurable supplier commitments.

2028 and later: Expand deployment based on risk and support. Track residual quantum-vulnerable paths, exceptions, replacements and standards updates.

Timeline interpretation: These horizons are suggested local planning phases. IR 8547 is draft. Algorithm-specific 2030/2035 proposals are not a universal mandate or a prediction of a quantum computer date.

Evidence workflow: Complete needs owner, evidence, review date and reviewer. Accepted Risk / Not Applicable also needs rationale and approval. Next action flags missing prerequisites.

Input legend: Amber cells are editable. P1/P2/P3 are local priorities, not NIST scores. Example inventory rows are fictional. Blank cells mean unassessed. No implementation results are asserted.

Update cadence: Recheck sources, vendor support, inventory evidence and exceptions quarterly and after material changes. This is a local cadence. Review new evidence before claiming alignment with future 2027+ guidance.

Controls: 38

Interpretation: Counts measure documentary prerequisites. They do not certify NIST compliance or independently validate implementation.

Checklist

Use each item with evidence, an accountable owner, a target date, and the editable workbook status fields. An unchecked control is not satisfied by this reference document.

01 Assign cloud/IAM PQC migration owner and define scope for cloud accounts, tenants, subscriptions, projects, and SaaS providers.

Domain: Governance

Why It Matters: Cloud and identity trust paths cross organizational boundaries and require accountable ownership.

Product-family mapping: Cloud services; SaaS; identity services

NIST/PQC Alignment: NIST NCCoE migration scope and inventory

Priority: P1

Control ID: CLD-C001

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

02 Define cryptographic change control for managed cloud services, identity providers, certificate authorities, service mesh, and CI/CD signing.

Domain: Governance

Why It Matters: Provider-managed services may change cryptographic behavior outside application release cycles.

Product-family mapping: Cloud services; web software

NIST/PQC Alignment: Crypto-agility governance

Priority: P1

Control ID: CLD-C002

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

03 Inventory all managed certificates, private CAs, public CAs, certificate automation tools, and TLS termination points.

Domain: Discovery

Why It Matters: Certificate and TLS dependencies are common PQC migration blockers.

Product-family mapping: Web software; cloud services

NIST/PQC Alignment: FIPS 203/204/205 planning; NCCoE discovery

Priority: P1

Control ID: CLD-C003

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

04 Inventory all workload identities: service principals, managed identities, IAM roles, Kubernetes service accounts, SPIFFE/SPIRE IDs, service mesh identities, and CI/CD identities.

Domain: Discovery

Why It Matters: Machine identities frequently rely on certificates, tokens, or signed assertions.

Product-family mapping: Cloud services; SaaS; identity services

NIST/PQC Alignment: Identity and key lifecycle planning

Priority: P1

Control ID: CLD-C004

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

05 Map TLS/mTLS endpoints by issuer, algorithm, key size, validity period, renewal automation, and termination layer.

Domain: Certificates

Why It Matters: PQC signature and key-establishment changes may affect certificate chain size and handshake behavior.

Product-family mapping: Web software; cloud services

NIST/PQC Alignment: FIPS 203 key establishment; FIPS 204/205 signatures

Priority: P1

Control ID: CLD-C005

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

06 Identify certificate chain-size, header-size, MTU, proxy, load balancer, API gateway, and service mesh limits.

Domain: Certificates

Why It Matters: PQC and hybrid credentials can increase message sizes and trigger interoperability issues.

Product-family mapping: Web software; cloud services

NIST/PQC Alignment: Interoperability testing

Priority: P1

Control ID: CLD-C006

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

07 Request CA roadmap for ML-DSA/SLH-DSA or supported hybrid certificate profiles where applicable.

Domain: PKI

Why It Matters: PKI migration depends on issuer and relying-party support.

Product-family mapping: Web software; cloud services

NIST/PQC Alignment: FIPS 204/205 signature planning

Priority: P1

Control ID: CLD-C007

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

08 Inventory token-signing algorithms for OIDC, OAuth, SAML, JWT, federation metadata, and session-token verification.

Domain: IAM

Why It Matters: Identity assurance can fail if relying parties cannot validate new signature algorithms.

Product-family mapping: SaaS; cloud services; identity services

NIST/PQC Alignment: Digital signature migration planning

Priority: P1

Control ID: CLD-C008

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

09 Map relying-party applications and libraries that validate tokens or signed assertions.

Domain: IAM

Why It Matters: Verifier compatibility is as important as issuer readiness.

Product-family mapping: SaaS; web software

NIST/PQC Alignment: Crypto-agility and interoperability testing

Priority: P1

Control ID: CLD-C009

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

10 Document Kubernetes/service-mesh certificate issuers, trust bundles, mTLS policies, rotation intervals, and sidecar/gateway constraints.

Domain: Workload Identity

Why It Matters: Workload identity often depends on short-lived certs and service-to-service trust.

Product-family mapping: Cloud services; web software

NIST/PQC Alignment: Certificate lifecycle and key establishment

Priority: P1

Control ID: CLD-C010

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

11 Document serverless, container, and VM identity mechanisms and whether they use signed tokens, certificates, cloud metadata, or workload attestation.

Domain: Workload Identity

Why It Matters: Provider-managed identity abstractions still depend on cryptographic primitives.

Product-family mapping: Cloud services

NIST/PQC Alignment: Inventory and supplier readiness

Priority: P2

Control ID: CLD-C011

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

12 Inventory KMS, HSM, secrets managers, envelope encryption, key wrapping, and external key store dependencies.

Domain: Key Management

Why It Matters: Key lifecycle controls must support migration without breaking availability or compliance.

Product-family mapping: Cloud services; endpoint/data-at-rest

NIST/PQC Alignment: Crypto-agility and C-SCRM

Priority: P1

Control ID: CLD-C012

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

13 Confirm key rotation, backup, escrow, destruction, and recovery processes remain valid during algorithm or provider changes.

Domain: Key Management

Why It Matters: Poorly sequenced key changes can break recovery or data access.

Product-family mapping: Cloud services; endpoint/data-at-rest

NIST/PQC Alignment: Operational resilience

Priority: P2

Control ID: CLD-C013

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

14 Inventory artifact signing, container signing, IaC signing, package repository signing, and build-provenance attestations.

Domain: CI/CD Signing

Why It Matters: Software supply-chain trust relies on signature verification at multiple points.

Product-family mapping: Cloud services; web software; endpoint security

NIST/PQC Alignment: FIPS 204/205 signature planning; SP 800-161

Priority: P1

Control ID: CLD-C014

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

15 Request SaaS providers' PQC readiness, certificate roadmap, token-signing roadmap, data-at-rest key lifecycle, and support-window commitments.

Domain: SaaS

Why It Matters: SaaS migration depends on supplier-controlled trust chains.

Product-family mapping: SaaS; cloud services

NIST/PQC Alignment: SP 800-161 supplier risk

Priority: P1

Control ID: CLD-C015

Planning horizon: 2026 baseline

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

16 Create non-production pilot for TLS/mTLS, identity token verification, API gateway, service mesh, logging, monitoring, and rollback.

Domain: Testing

Why It Matters: PQC readiness should be demonstrated before enterprise rollout.

Product-family mapping: Cloud services; web software

NIST/PQC Alignment: NCCoE interoperability testing

Priority: P1

Control ID: CLD-C016

Planning horizon: 2027 pilots and rollout

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

17 Update runbooks for certificate renewal failure, token-signing key changes, trust-bundle distribution, emergency rollback, and vendor outage.

Domain: Operations

Why It Matters: Migration failure modes require operational recovery paths.

Product-family mapping: Cloud services; web software; SaaS

NIST/PQC Alignment: Operational continuity

Priority: P2

Control ID: CLD-C017

Planning horizon: 2027 pilots and rollout

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

18 Add PQC support, crypto inventory/CBOM, roadmap, testing evidence, and notification clauses to vendor evaluations.

Domain: Procurement

Why It Matters: Future procurement should not introduce unmanaged quantum-vulnerable dependencies.

Product-family mapping: CISA product categories; cloud services; web software; endpoint security

NIST/PQC Alignment: SP 800-161 C-SCRM

Priority: P1

Control ID: CLD-C018

Planning horizon: 2027 pilots and rollout

Guidance status: Local task mapped to NIST guidance

Threat / gap: See task rationale

19 Record confidentiality lifetime and capture exposure for traffic, backups, secrets exchange and retained sensitive data.

Domain: Threat assessment

Why It Matters: Harvest-now-decrypt-later risk depends on data lifetime and capture exposure.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N06, N07

Priority: P1

Control ID: CLD-C019

Planning horizon: 2026 baseline

Guidance status: Includes draft watch; see Sources

Threat / gap: Harvest now, decrypt later

20 Record verification lifetime for tokens, trust anchors, release artifacts, audit evidence and archived records.

Domain: Threat assessment

Why It Matters: Future forgery affects authentication and long-lived authenticity differently from confidentiality.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N02, N03, N09, N16

Priority: P1

Control ID: CLD-C020

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Future signature forgery

21 Record library/provider, exact build, protocol version, parameter set, config location, discovery method, last observed date and gaps.

Domain: Discovery

Why It Matters: Product names and account inventories alone do not identify cryptographic use.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N07, N08

Priority: P1

Control ID: CLD-C021

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Hidden or stale dependencies

22 Map issuers, intermediaries, all relying parties, partner APIs, multi-region paths, egress, backups and recovery environments.

Domain: Discovery

Why It Matters: The least-ready trust dependency can block migration.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N07, N16

Priority: P1

Control ID: CLD-C022

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Unmapped trust and recovery paths

23 Separate ML-KEM shared-secret establishment from ML-Dsa/SLH-Dsa signatures. Record parameter set, KDF and consuming protocol.

Domain: Algorithm selection

Why It Matters: KEM adoption does not replace certificate or token signatures.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N01, N02, N03, N04

Priority: P1

Control ID: CLD-C023

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Wrong primitive or parameters

24 Review vetted implementation, randomness, key separation, input validation, decapsulation failure behavior and shared-secret destruction.

Domain: KEM engineering

Why It Matters: Correct algorithms can fail through implementation or key-management defects.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N04, N09

Priority: P1

Control ID: CLD-C024

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Malformed ciphertext, side channels or secret leakage

25 Verify CMVP certificate, module/version, operating environment, approved mode, PQC service coverage and current status where required.

Domain: Module assurance

Why It Matters: CAVP or a marketing claim is not module, protocol or whole-product validation. Historical is not the same as revoked.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N24, N25, N26

Priority: P1

Control ID: CLD-C025

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Unsupported assurance claim

26 Record protocol profile and maturity for TLS, SSH, VPN, JOSE/COSE, OIDC/SAML and X.509. Test exact endpoint/verifier versions.

Domain: Interoperability

Why It Matters: Final algorithms do not make every protocol integration standardized or supported.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N01, N02, N03, N08, N16

Priority: P1

Control ID: CLD-C026

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Protocol/profile incompatibility

27 Document supported hybrid construction, combiner, negotiation and downgrade policy. Test failure of either component and log negotiated results.

Domain: Hybrid policy

Why It Matters: Hybrid properties depend on protocol construction and implementation.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N04, N05

Priority: P1

Control ID: CLD-C027

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Downgrade or insecure composition

28 Measure handshake, certificate and token sizes, HTTP header/cookie limits, CPU, RAM, concurrency, tail latency, MTU and denial-of-service behavior.

Domain: Capacity

Why It Matters: Larger keys, signatures and messages can disrupt authentication or availability.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N08, N16, N17

Priority: P1

Control ID: CLD-C028

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Resource exhaustion or availability loss

29 Test JWKS/metadata distribution, caches, key identifiers, rollover overlap, token expiry, and rejection of disallowed algorithms.

Domain: Federation

Why It Matters: Successful issuance does not prove safe verification across relying parties.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N09, N16

Priority: P1

Control ID: CLD-C029

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Stale trust or algorithm confusion

30 Include service and automated workload signed-token issuers and every validator. Verify audience, issuer, lifetime and delegated scope.

Domain: Workload identity

Why It Matters: A new token signature cannot repair excessive privileges or incorrect token validation.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N16, N17, N22, N23

Priority: P1

Control ID: CLD-C030

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Token replay, excessive delegation or validator gaps

31 Test enrollment, renewal, revocation, chain building and trust-store distribution across all supported issuer/verifier versions.

Domain: PKI lifecycle

Why It Matters: A migrated CA can still break trust, renewal or revocation.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N02, N03, N09

Priority: P1

Control ID: CLD-C031

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Trust outage or stale revoked credentials

32 Inventory archive/backup key wrapping. Test old-data restore, key recovery and rewrapping. Assess bulk symmetric encryption separately.

Domain: Data recovery

Why It Matters: Long-lived data depends on protecting its key management and recovery paths.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N04, N09

Priority: P1

Control ID: CLD-C032

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Loss of confidentiality or recovery

33 Require exact support versions, release dates, interoperability evidence, security updates, lifecycle dates and notification commitments.

Domain: Supplier evidence

Why It Matters: A roadmap without version-level evidence cannot support an approved wave.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N29

Priority: P1

Control ID: CLD-C033

Planning horizon: 2026 baseline

Guidance status: Local task; source status in Sources

Threat / gap: Supplier delay or unsupported products

34 Record owner-approved tests, monitoring, incident contacts and rollback. Obtain production authorization separately from a successful lab result.

Domain: Release approval

Why It Matters: Deployment decisions require complete operational evidence.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N27

Priority: P1

Control ID: CLD-C034

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Premature rollout or outage

35 Record unsupported dependencies, compensating controls, approver, review/expiry date and replacement milestone.

Domain: Exception management

Why It Matters: Accepted residual risk does not make a legacy path quantum resistant.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N27

Priority: P1

Control ID: CLD-C035

Planning horizon: 2027 pilots and rollout

Guidance status: Local task; source status in Sources

Threat / gap: Permanent legacy exception

36 Rediscover cryptography after releases, acquisitions, provider changes and trust updates. Detect classical fallback and inventory drift.

Domain: Continuous assurance

Why It Matters: Crypto agility requires maintained evidence after initial migration.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N07

Priority: P1

Control ID: CLD-C036

Planning horizon: 2028+ continuous migration

Guidance status: Local task; source status in Sources

Threat / gap: Silent regression

37 Review FIPS errata, IR 8547, draft SP 800-131A Rev. 3, SP 800-57 Part 1 Rev. 6, SP 800-133 Rev. 3 and SP 800-230; monitor pending FN-DSA/HQC.

Domain: Standards watch

Why It Matters: Drafts, candidates and forecasts are not finalized approvals. Do not defer current migration for a future algorithm.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N01-N06, N10, N12, N19-N21, N30, N31

Priority: P1

Control ID: CLD-C037

Planning horizon: 2028+ continuous migration

Guidance status: Includes draft watch; see Sources

Threat / gap: Stale guidance or premature standards claims

38 Retire replaced keys, trust anchors, configs and unsupported dependencies after validating recovery and retention obligations.

Domain: Retirement

Why It Matters: Migration is incomplete while unmanaged fallback or verification remains.

Product-family mapping: Cloud / identity / supply chain

NIST/PQC Alignment: N05, N09

Priority: P1

Control ID: CLD-C038

Planning horizon: 2028+ continuous migration

Guidance status: Local task; source status in Sources

Threat / gap: Residual quantum-vulnerable paths

Workload Identity Inventory

The editable workbook contains the full data-entry template. The field structure below identifies information to collect. Example records are illustrative and are not a validated production inventory.

Fields in the editable worksheet

01. Platform
02. Workload/App
03. Environment
04. Identity Type
05. Token/Certificate Mechanism
06. Signing Algorithm
07. Issuer / Trust Anchor
08. Rotation Period
09. External Exposure
10. PQC Dependency
11. Migration Blocker
12. Priority
13. Owner
14. Evidence Link
15. Notes
16. Asset / flow ID

17. Crypto library/provider version
18. Protocol/profile and version
19. Key establishment / KEM
20. Parameter set
21. Confidentiality lifetime (years)
22. Signature verification lifetime
23. Discovery method
24. Last observed
25. Supplier/version evidence
26. CMVP evidence if required
27. Migration wave
28. Exception approver / review date

Illustrative records - selected fields

01 Record TLS/mTLS profile

Signing Algorithm: ECDSA/RSA

Notes: Example only

Asset / flow ID: EX-CLD-001

02 Record artifact signature profile

Signing Algorithm: RSA

Notes: Example only

Asset / flow ID: EX-CLD-002

Certificate and Key Map

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

Fields in the editable worksheet

01. Endpoint / Service
02. TLS/mTLS/Signing Use
03. Issuer
04. Algorithm
05. Key Size / Curve
06. Validity Period
07. Automation Tool
08. Termination Layer
09. Relying Parties
10. PQC Roadmap
11. Pilot Needed?
12. Priority
13. Evidence Link
14. Notes
15. Asset / flow ID
16. Key establishment algorithm
17. Signature parameter set
18. Profile/status and support version
19. Chain / token bytes
20. KMS/HSM module and CMVP evidence
21. Expiry date
22. Revocation / recovery evidence
23. All verifier versions
24. Last verified

25. Approved change / exception

Illustrative records - selected fields

01 Unknown: observe handshake

Algorithm: RSA/ECDSA

Notes: Example only

Asset / flow ID: EX-CLD-001

02 Unknown: observe handshake

Algorithm: ECDSA

Notes: Example only

Asset / flow ID: EX-CLD-004

CISA Category Map

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

01 IaaS, PaaS, managed TLS, KMS/HSM, secrets, API gateways

Product family (local map): Cloud services

Checklist Focus: Managed crypto, certificates, workload identity, KMS/HSM, provider roadmap

Readiness Evidence to Request: PQC support statement, API/configuration guidance, test evidence, support timeline

Migration Concern: Provider-managed cryptography can constrain enterprise sequencing.

Support status: Unverified

02 Browsers, web servers, proxies, WAF, load balancers

Product family (local map): Web software

Checklist Focus: TLS, certificate chain size, protocol negotiation, monitoring, rollback

Readiness Evidence to Request: Supported protocol versions, PQC/hybrid documentation, interoperability tests

Migration Concern: Client/server mismatch can break availability.

Support status: Unverified

03 Endpoint security / data-at-rest

Examples: Full disk encryption, data-at-rest security, device protection

Checklist Focus: Long-lived data, key wrapping, KMS, recovery process

Readiness Evidence to Request: Algorithm disclosure, key lifecycle process, recovery test

Migration Concern: Data recovery and device management must remain reliable.

Support status: Unverified

04 Collaboration software

Examples: Chat, messaging, collaboration platforms

Checklist Focus: Vendor PQC roadmap, retention, eDiscovery/export, identity integration

Readiness Evidence to Request: Vendor roadmap, encryption architecture, support timeline

Migration Concern: SaaS-controlled cryptography and long-lived retained data.

Support status: Unverified

05 Identity and access management

Examples: IdP, federation, privileged access, machine identities

Checklist Focus: Token signing, federation metadata, relying-party verification

Readiness Evidence to Request: Algorithm roadmap, metadata rollout procedure, verifier compatibility

Migration Concern: Identity token verification failures can cause broad outages.

Support status: Unverified

06 CI/CD and artifact signing

Examples: Build systems, package repositories, container signing, release signing

Checklist Focus: Signature algorithm, key protection, verifier support

Readiness Evidence to Request: Signing roadmap, verifier matrix, rollback plan

Migration Concern: Build trust chain can fail if verifiers are not migrated.

Support status: Unverified

Pilot Test Plan

Capture a baseline and explicit acceptance criteria, test in the approved environment, and preserve result and rollback evidence before authorizing a rollout.

01 TLS/mTLS pilot through API gateway and service mesh

Test ID: CLD-001

Control Point: Gateway, proxy, service mesh

Metric: Handshake success, latency, error rate, logging

Acceptance Criteria: No critical errors; latency within approved threshold; rollback tested.

NIST sources: N04, N05, N08, N09, N16, N17

02 OIDC/JWT token-signing compatibility test

Test ID: CLD-002

Control Point: IdP and relying parties

Metric: Token validation success/failure, library support

Acceptance Criteria: All in-scope relying parties pass expected token-validation tests. Record failures, exclusions and risk approvals separately. An exception is not a test pass.

NIST sources: N04, N05, N08, N09, N16, N17

03 Artifact signing verification test

Test ID: CLD-003

Control Point: CI/CD and package consumers

Metric: Signature creation/verification, artifact deployment success

Acceptance Criteria: Build and deployment pipeline succeeds; failed signature blocks deployment.

NIST sources: N04, N05, N08, N09, N16, N17

04 KMS/HSM key lifecycle drill

Test ID: CLD-004

Control Point: KMS/HSM/secrets

Metric: Rotation, backup, recovery, audit logs

Acceptance Criteria: Recovery and rotation procedure passes with evidence.

NIST sources: N04, N05, N08, N09, N16, N17

05 Rollback and incident runbook exercise

Test ID: CLD-005

Control Point: Operations

Metric: Recovery time, decision path, evidence capture

Acceptance Criteria: Rollback completes within approved window and records are retained.

NIST sources: N04, N05, N08, N09, N16, N17

06 Malformed KEM ciphertext and unsupported parameters

Test ID: CLD-006

Control Point: KEM library/provider

Metric: Failure behavior and secret handling

Acceptance Criteria: No crash, secret disclosure or silent fallback. Approved validation and failure behavior.

NIST sources: N04, N05, N08, N09, N16, N17

07 Hybrid negotiation and downgrade attempt

Test ID: CLD-007

Control Point: Supported TLS/VPN/SSH profile

Metric: Negotiated algorithms and component failures

Acceptance Criteria: Behavior matches policy. Downgrade is rejected or explicitly approved and logged.

NIST sources: N04, N05, N08, N09, N16, N17

08 Large certificate and token path

Test ID: CLD-008

Control Point: WAF, proxy, gateway and clients

Metric: Bytes, headers, cookies, MTU and tail latency

Acceptance Criteria: All supported paths satisfy recorded owner-approved limits.

NIST sources: N04, N05, N08, N09, N16, N17

09 Concurrent handshake and failure storm

Test ID: CLD-009

Control Point: Mesh and identity endpoints

Metric: CPU, RAM, handshake rate and errors

Acceptance Criteria: Peak-load results meet recorded limits and rate limiting preserves service.

NIST sources: N04, N05, N08, N09, N16, N17

10 JWKS, metadata and token rollover

Test ID: CLD-010

Control Point: IdP and every validator

Metric: Cache refresh, expiry, audience and issuer checks

Acceptance Criteria: Valid tokens work; invalid, expired or disallowed-algorithm tokens fail.

NIST sources: N04, N05, N08, N09, N16, N17

11 Historical backup and key recovery

Test ID: CLD-011

Control Point: KMS/HSM and backup platform

Metric: Restore time and data integrity

Acceptance Criteria: Restore with approved keys meets owner-approved recovery target.

NIST sources: N04, N05, N08, N09, N16, N17

12 CA renewal and revocation lifecycle

Test ID: CLD-012

Control Point: CA, agents and verifiers

Metric: Enrollment, renewal, chain and revocation

Acceptance Criteria: Each supported verifier handles lifecycle states according to policy.

NIST sources: N04, N05, N08, N09, N16, N17

13 Old artifact verification and tamper rejection

Test ID: CLD-013

Control Point: CI/CD and offline consumers

Metric: Signature validation and release controls

Acceptance Criteria: Retained artifacts remain verifiable. Tampering blocks release.

NIST sources: N04, N05, N08, N09, N16, N17

14 Regional failover and provider outage

Test ID: CLD-014

Control Point: Cloud recovery path

Metric: Authentication, failover time and logging

Acceptance Criteria: Recovery identity paths meet recorded security and availability limits.

NIST sources: N04, N05, N08, N09, N16, N17

15 Library update and inventory drift

Test ID: CLD-015

Control Point: Discovery and release pipeline

Metric: Version detection and evidence refresh

Acceptance Criteria: Changes are detected and assessed before release; inventory matches deployed version.

NIST sources: N04, N05, N08, N09, N16, N17

Sources

Official sources, publication status, and applicability recorded in the workbook. Recheck draft status, errata, and implementation guidance before a production decision.

01 FIPS 203 ML-KEM; key establishment, parameter sets 512/768/1024; monitor potential errata

Source ID: N01

Status as of 2026-09-29: Final 2024-08-13; notice 2025-11-17

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/fips/203/final>

02 FIPS 204 ML-DSA; signature generation and verification; monitor potential errata

Source ID: N02

Status as of 2026-09-29: Final 2024-08-13; notice 2026-07-31

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/fips/204/final>

03 FIPS 205 SLH-DSA; stateless hash-based signatures

Source ID: N03

Status as of 2026-09-29: Final 2024-08-13

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/fips/205/final>

04 SP 800-227; conforming KEM implementation, randomness, input validation, key destruction, authentication, key derivation/confirmation and multi-algorithm construction

Source ID: N04

Status as of 2026-09-29: Final 2025-09-18

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/227/final>

05 CSWP 39-upd1; governance, inventory, risk-prioritized agility, negotiation, supply chains, APIs, continuous improvement

Source ID: N05

Status as of 2026-09-29: Final 2025-12-19; updated 2026-06-29

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/cswp/39/upd1/considerations-for-achieving-crypto-agility/final>

06 IR 8547; HNDL, signature and code-verification distinctions, draft transition tables

Source ID: N06

Status as of 2026-09-29: Initial public draft 2024-11-12

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/ir/8547/ipd>

07 NCCoE Migration to PQC; discovery/inventory and interoperability workstreams; evolving output model

Source ID: N07

Status as of 2026-09-29: Active project; preliminary documents

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://www.nccoe.nist.gov/applied-cryptography/migration-to-pqc>

08 SP 1800-38 B/C; discovery and interoperability/performance demonstration

Source ID: N08

Status as of 2026-09-29: Initial preliminary draft 2023-12-19

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: [https://csrc.nist.gov/pubs/sp/1800/38/iprd-\(1\)](https://csrc.nist.gov/pubs/sp/1800/38/iprd-(1))

09 SP 800-57 Part 1 Rev. 5; key management baseline

Source ID: N09

Status as of 2026-09-29: Final 2020-05-04

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final>

10 SP 800-57 Part 1 Rev. 6; PQC-aware key management and storage proposals

Source ID: N10

Status as of 2026-09-29: Initial public draft 2025-12-05

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r6/ipd>

11 SP 800-131A Rev. 2; current final algorithm transition baseline

Source ID: N11

Status as of 2026-09-29: Final 2019-03-21

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/131/a/r2/final>

12 SP 800-131A Rev. 3; proposed algorithm/key-strength transitions

Source ID: N12

Status as of 2026-09-29: Initial public draft 2024-10-21

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/131/a/r3/ipd>

13 SP 800-82 Rev. 3; OT performance, reliability, safety and controls

Source ID: N13

Status as of 2026-09-29: Final 2023-09-28

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

14 SP 800-82 Rev. 4; CSF2 alignment, asset management, monitoring, OT/IIoT/cloud, system-management security

Source ID: N14

Status as of 2026-09-29: Initial public draft 2026-09-21

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/82/r4/ipd>

15 SP 1339; backups in OT change and recovery practice

Source ID: N15

Status as of 2026-09-29: Final 2026-06-17

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/1339/final>

16 IR 8587; signed tokens/assertions, key lifecycle, verification, workload/automated identity token use, Section 7.7 PQC interoperability constraints

Source ID: N16

Status as of 2026-09-29: Final 2026-09-15

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/ir/8587/final>

17 SP 800-228-upd1; API lifecycle risks and controls

Source ID: N17

Status as of 2026-09-29: Final; original 2025-06-27, updated 2026-03-13

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/228/upd1/final>

18 CSWP 48; mapping PQC discovery and interoperability capabilities to risk frameworks

Source ID: N18

Status as of 2026-09-29: Initial public draft 2025-09-18

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/cswp/48/mapping-migration-to-pqc-project-capabilities-to-r/tpd>

19 NIST PQC project; current standardized/selected algorithm status

Source ID: N19

Status as of 2026-09-29: Project updated 2026-08-05

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/post-quantum-cryptography>

20 HQC selection announcement; diversification backup, continue current migration, expected 2027 finalization

Source ID: N20

Status as of 2026-09-29: Selection announcement 2025-03-11, forecast not approval

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://www.nist.gov/news-events/news/2025/03/nist-selects-hqc-fifth-algorithm-post-quantum-encryption>

21 Additional digital signature schemes; Round3 announced May14,2026

Source ID: N21

Status as of 2026-09-29: Evaluation candidates, not standards

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/pqc-dig-sig>

22 SP 800-204A; service-mesh security architecture

Source ID: N22

Status as of 2026-09-29: Final 2020-05-27

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/204/a/final>

23 SP 800-204B; service-mesh authentication/authorization

Source ID: N23

Status as of 2026-09-29: Final 2021-08

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/204/b/final>

24 CAVP program; algorithm validation is prerequisite to, not replacement for, module validation

Source ID: N24

Status as of 2026-09-29: Official program; updated 2026-09-22

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program>

25 CMVP FAQ; module/version/operating-environment verification, protocol/product validation boundary, FIPS140-2 active-list transition

Source ID: N25

Status as of 2026-09-29: Official FAQ checked 2026-09-29

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/Projects/cryptographic-module-validation-program/faqs>

26 CMVP validated modules; inspect certificate/security policy and active/historical/revoked status

Source ID: N26

Status as of 2026-09-29: Live certificate registry checked 2026-09-29

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/cryptographic-module-validation-program/validated-modules>

27 CSWP 29 / CSF 2.0; cybersecurity outcome framework, not a prescriptive PQC implementation standard

Source ID: N27

Status as of 2026-09-29: Final 2024-02-26

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/cswp/29/the-nist-cybersecurity-framework-csf-20/final>

28 SP 800-53 Rev. 5; current control catalog is Release 5.2.0 referenced by publication-page planning note

Source ID: N28

Status as of 2026-09-29: Final; base 2020-09/update 2020-12-10; minor Release 5.2.0 finalized 2025-08-27

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final>

29 SP 800-161 Rev. 1-upd1; supplier risk and dependency governance

Source ID: N29

Status as of 2026-09-29: Final; original 2022-05, update 2024-11-01

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/161/r1/upd1/final>

30 SP 800-230; additional limited-signature SLH-DSA parameter sets, software/firmware/certificate watch item

Source ID: N30

Status as of 2026-09-29: Initial public draft 2026-04-13

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/230/ipd>

31 SP 800-133 Rev. 3; proposed PQC key generation, seed expansion, KEM and HSM considerations

Source ID: N31

Status as of 2026-09-29: Initial public draft 2026-04-17

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/133/r3/ipd>

32 CISA - Product Categories for Technologies That Use Post-Quantum Cryptography Standards

Source ID: C1

Status as of 2026-09-29: Supplementary guidance; see publication

Interpretation: 2026 product-category reference for PQC adoption, including cloud services, web software, and endpoint security categories.

Public URL: <https://content.govdelivery.com/accounts/USDHSCISA/bulletins/405c246>

33 CISA/NSA/NIST - Quantum-Readiness: Migration to Post-Quantum Cryptography

Source ID: C2

Status as of 2026-09-29: Supplementary guidance; see publication

Interpretation: Quantum-readiness roadmap, cryptographic inventory, vendor engagement, and supply-chain dependency assessment.

Public URL:

<https://www.cisa.gov/resources-tools/resources/quantum-readiness-migration-post-quantum-cryptography>

34 CISA - Post-Quantum Considerations for Operational Technology

Source ID: C3

Status as of 2026-09-29: Supplementary guidance; see publication

Interpretation: OT/ICS/SCADA migration considerations for safety, performance, vendor dependency, and operational continuity.

Public URL: <https://www.cisa.gov/resources-tools/resources/post-quantum-considerations-operational-technology>